

Mohamed Hassan Ismail Elsis

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CAREER OBJECTIVE

Junior Machine Learning Engineer in training through the DEPI Microsoft ML Engineer track, covering 6+ core domains including Python, machine learning, deep learning, NLP, computer vision, and Microsoft Azure AI. A collaborative, detail-oriented team member with hands-on experience contributing to 1 comprehensive embedded systems project. Seeking to apply strong analytical and problem-solving fundamentals, alongside a growing machine learning skill set, to an entry-level ML/AI role.

EDUCATION

Misr University for Science and Technology (MUST) — Egypt | Expected 2027

Faculty of Information Technology, Artificial Intelligence Major — GPA: 3.57 / 4.0

EXPERIENCE

Machine Learning Trainee | Jul. 2026 – Present

Digital Egypt Pioneers Initiative (DEPI), Microsoft ML Engineer Track — Remote

- Enrolled in an 8-module, 120-hour project-based curriculum spanning a 6-month track, maintaining a rigorous pace by averaging 20+ hours of structured coursework per month.
- Accelerated learning progress by completing 2 of 8 modules (mathematics, Python) within the first 30 days, successfully finishing 25% of the full curriculum in month 1 alone.
- Engineered data preprocessing pipelines for 2 real-world datasets totaling 21,000+ combined records (Titanic, California Housing), handling 100% of data cleaning and feature engineering tasks.
- Utilized 4 core Python libraries (NumPy, Pandas, Matplotlib, Seaborn) to generate 10+ exploratory data analysis (EDA) visualizations, extracting actionable insights and identifying key correlations.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, SQL
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn
- **Machine Learning (in progress):** Scikit-learn, supervised and unsupervised learning, regression, classification, clustering, model evaluation
- **Deep Learning & AI (in progress):** TensorFlow, neural networks, NLP, Computer Vision
- **Cloud & MLOps (in progress):** Microsoft Azure AI, Azure Machine Learning, MLFlow
- **Core CS Foundations:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Systems (SQL, ER modeling)
- **Tools:** VS Code, Jupyter Notebook, Google Colab, Git, Arduino IDE, Postman
- **Databases & BI:** MySQL, SQL Server, Excel, Power BI
- **Languages:** Arabic (Native), English (Fluent)

PROJECTS

Minimum Coins Payment System | C++, Dynamic Programming

- Optimized algorithm efficiency by reducing time complexity from exponential (brute-force recursion) to polynomial $O(n \times A)$, resulting in an estimated 90%+ decrease in processing time for large denomination inputs.
- Achieved a 100% validation success rate across 5+ worked examples and 10+ edge cases, rigorously benchmarking the DP solution against the greedy algorithm.

- Coordinated within a 4-member cross-functional team to deliver a comprehensive 3-part technical report (pseudocode, flowchart, Greedy vs. DP comparison), fulfilling 100% of the project requirements with meticulous attention to detail.
- Documented and maintained $O(n \times A)$ time complexity and $O(A)$ space complexity specifications within the C++ implementation.

CERTIFICATIONS

- **Machine Learning Engineer Track (In Progress)** — Digital Egypt Pioneers Initiative (DEPI), Microsoft